

UNIT- 2

Practice – 5

1. File Manipulation:

- List the contents of your current directory using `ls`.
- Change directory to a different location using `cd`.
- Create a new empty file named `data.txt` using `touch`.
- View the contents of `data.txt` using `cat`.

2. Redirection Operators:

- Append the following line to `data.txt` using `echo "This is some sample data" >> data.txt`.
- Overwrite the contents of `data.txt` with the same line using `echo "This is some sample data" > data.txt`.
- Redirect the standard output of the `ls` command to a new file named `filelist.txt` using `ls > filelist.txt`.

3. Standard Streams (STDIN, STDOUT, STDERR):

- Explain the difference between Standard Input (STDIN), Standard Output (STDOUT), and Standard Error (STDERR).
- Provide an example of how each is used in a shell command.

4. Grep Command:

- Use `grep "data" on data.txt` to find lines containing the word "data".
- Search for lines that **don't** contain "data" using `grep -v "data" data.txt`.
- Combine `grep` with pipes (`|`) to search for lines with "data" and then filter for those containing "sample" using `grep "data" data.txt | grep "sample"`.

5. Cut Command:

- Use `cut -d" " -f2 on data.txt` to extract the second word (separated by spaces) from each line.
- Modify the command to extract a specific range of characters (e.g., first 5 characters) using `cut -c 1-5 data.txt`.

6. Awk Command:

- Print only the second field from `data.txt` using `awk '{print $2}' data.txt`.
- Count the number of lines in `data.txt` with `awk END {print NR}' data.txt` (NR refers to the number of records).

7. Command Chaining:

- Combine `grep`, `cut`, and `awk` to find lines containing "data", extract the second word, and print only the first three characters using a single command with pipes.